Kanghyun Ko

Backend Developer

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I'm Kanghyun Ko, a developer with a broad perspective. At startups, I have worked from planning through launch and operations, gaining experience in business, marketing and product planning. This taught me to draw on how people in other fields frame and solve problems. Engineering blog discussions have helped me learn how senior engineers reason about problems and make technical decisions.

With this broader outlook, **I identify problems in everyday life and services, and adapt solutions from different experiences to the situation at hand.** I continue to broaden my perspective through engineering blog study groups and cloud and IT communities. I apply what I learn to personal projects addressing everyday frustrations, refining solutions through direct use and feedback.

Skills

Backend	TypeScript, Node.js, NestJS, Python, FastAPI, BullMQ, WebSocket
Data	PostgreSQL, MongoDB, Redis, Qdrant
Frontend	React, Next.js
Infrastructure	AWS EC2·S3·Lambda, Docker, GitHub Actions, Sentry

Experience

Full-time: 1 year · Development: 2 years 2 months

Grapic

Aug 2025 - Sep 2025

Part-time · **Full-stack Developer**

Python, FastAPI, AWS Lambda, TypeScript, Next.js, React, Docker

Refund-aware revenue reporting: Incomplete monthly reporting and refund handling required manual revenue calculations → discussed improvements with operators and designers and connected payment gateway refund records → enabled revenue reporting that accounts for refunds.

Community-specific crawlers: Built a class-based structure around shared crawling logic, allowing collection logic for different community sites to be applied and reused.

On-demand crawling: Built FastAPI-based crawling and API servers, with AWS Lambda executing the crawling logic on request to lower operating costs.

Hashscape

Jul 2024 - Jul 2025

Solthumb

Multichain token trading and asset management

Feb 2025 - Jul 2025

Full-time · **Founding Team / Full-stack Developer**

TypeScript, NestJS, MongoDB, Redis, BullMQ, React Native

External API cost reduction

- Reliance on multiple APIs providing parsed transaction data drove costs above KRW 1.2 million → canceled those subscriptions and retained an API for raw data lookup by transaction ID → moved parsing, normalization and storage into the service.
- Separated chain-specific collection, parsing and storage stages and introduced a common data structure to support additional chains.

Transaction processing and balance consistency

- Transactions were confirmed on-chain before external APIs reflected them, causing failed lookups → changed the request flow and introduced BullMQ reprocessing to retrieve and process delayed results.
- Recorded balance adjustments and transaction history before execution, then maintained status and records at each stage to reduce lost transaction records. Restored amounts excluding gas fees on final failure.
- PnL calculation errors and chain-specific fees caused balance discrepancies → exposed queue states and error records in the admin tool for diagnosis → built recovery that reads actual on-chain balances and synchronizes the service database.

Trading fees and referrals

- **EVM:** High, non-configurable third-party fees conflicted with the service's low-fee offering → developed and deployed contracts across EVM chains, including fee collection and distribution logic → replaced external trading APIs and reduced costs.
- **EVM events:** Included required transaction data in custom contract events to simplify parsing, standardize data and reduce reliance on external APIs.
- **Solana:** Integrated a third-party feature that allocates part of transaction fees to a designated recipient, enabling referral fee collection.

Real-time charts

- Initial data retrieval and real-time connection setup introduced waiting time → reused Redis data fetched within the previous minute to reduce repeated initial-data requests.
- One-minute polling lacked real-time responsiveness → switched to WebSocket trade execution data, filtered abnormal prices and updated charts with valid trades; built an order-book-style display from executed trades.

Hedera data integration: No suitable transaction lookup API was available → requested foundation support and collaborated with an external company to secure and integrate API access.

> **Solthumb received a development grant from the Hedera foundation**

Salt Hub Telegram Mini App for initial user acquisition Jul 2024 - Feb 2025

Full-time · **Founding Team / Full-stack Developer**

TypeScript, NestJS, Next.js, MongoDB, Redis, Cloudflare, AWS · Telegram Mini App

Shake game and ticket settlement

- Implemented the full game, awarding points based on smartphone gyroscope shake counts. Supported five daily plays per user and boosters that consume multiple tickets at once.
- Upfront ticket deduction caused losses when users exited; concurrent games could reuse one ticket → introduced per-game identifiers, token validation and renewal, and separated preliminary Redis deductions from final database deductions → deducted tickets and awarded points only on completion, preventing duplicate settlement.
- Backgrounding and network delays caused token expiry → restored preliminary Redis deductions on abnormal termination, leaving database tickets undeducted for incomplete games.

Multiregion lookup failures: Replication lag caused new-user lookups to fail across domestic and global deployments → compared database logs and write timestamps to identify the cause → waited for majority write acknowledgment before subsequent lookups.

Request latency: Observed delays along the Cloudflare edge proxy path → migrated to an AWS-based configuration to improve latency.

> **400,000+ cumulative users · 2,000–3,000 monthly active users**

Newweb Communication

Jul 2023 - Jun 2024

Freelance · **Automation & Web Server Developer**

Next.js, NestJS, Vercel, Python

Marketing preparation and publishing automation

- Manually running separate programs each day required more than two hours of marketing preparation.
- Integrated image collection, video and thumbnail creation, text generation and publishing tools, with parallel execution to reduce repetitive preparation.
- Combined image color adjustment and video creation, with slicing and reuse of collected images to connect the content production workflow.
- Standardized titles and content in an Excel template: placing the file in a folder triggered publishing. Configured Windows and Bash to automate the previously manual daily execution.

Platform planning and development: Planned and built an MVP connecting clients with lawyers for legal representation.

> **Marketing preparation: >2 hours → <30 minutes · 75%+ reduction**

Inspien

Mar 2023 - Jun 2023

Internship · **Backend Developer**

Java, Spring Boot, Thymeleaf, MariaDB

Data integration: Consolidated data from distributed legacy and new-service databases.

Billing workflows: Built internal management screens and invoice email functionality for customer billing.

Service research: Researched new EDI services.

Activities

Cloud Club

Cloud engineering community

Sep 2026 - Present

Technical study: Studying Designing Data-Intensive Applications (DDIA) and large-scale data processing with Flink.

Engineering Blog Study Group

Studying and sharing industry engineering cases

Jan 2026 - Present

Learning and sharing: Study and discuss engineering blog posts from companies including Toss, Daangn and Olive Young.

Nexters · Logit AI application essay service based on personal experience Jan 2026 - Mar 2026

IT project community · **Backend Lead & Server Developer** · **Currently operating** logit.ai.kr

Python, FastAPI, PostgreSQL, Redis, Qdrant, FastMCP, Next.js, Docker, Caddy

MCP: User interviews revealed concerns about an additional AI subscription → proposed and built 10 MCP tools for saved experience and application essay data → enabled use of Logit data from existing ChatGPT and Claude workflows.

RAG: Needed to select relevant experience for a question, role and company → combined tag similarity and competency matching with RAG retrieval → implemented context-aware experience recommendations for application essays.

Admin

- Operators had to query dev/prod databases and vector stores separately → built an admin tool combining users, projects, questions, experiences and chats across both environments → unified data lookup and management.
- Following payment gateway registration, collected and managed token usage data to assess and adjust the business model while maintaining the service.

Deployment and error tracking: Set up dev/prod deployment environments, 59 automated tests and Sentry error tracking. Completed payment gateway registration and continue to operate the service.

Software Maestro, 13th Cohort · Never Jun 2022 - Dec 2022

Automatic music tagging and search iOS app using Naver Music data

Software talent development program · **Team Lead & Server/AI Developer**

Python, Django, TensorFlow, pandas, Elasticsearch · iOS app

Automatic tagging: Visualized audio as time-series images and compared them with reference mood-tag images. Used text character-count comparisons and similarity checks against emotion and situation tags to generate tags.

Tag and music search: Implemented similarity-based tag and music search with Elasticsearch. Delivered the project's final presentation at Naver.

Personal Projects

Web app

Tudini Game-based investment learning

Kotlin, Spring Boot, Kafka, Kubernetes, React, Docker

- A simulated trading service under development to lower the barrier to investment learning while studying trading systems and large-scale data processing. Configuring investors with different trading preferences in Kubernetes pods and adjusting their numbers to address low initial activity; developing news-responsive trading and price changes. Targeting launch as a Toss in-app service.

Web service

Jobgil Integrated job preparation jobgil.com

TypeScript, NestJS, Next.js, PostgreSQL(Supabase), Docker

- Connects job collection, company/role analysis, resume feedback, application tracking and morning notifications to reduce fragmented preparation across tools. Used personally for about six months; deployed to around 10 users and improved through their feedback.

macOS desktop app

DevMate Integrated AI development environment

TypeScript, Bun, Tauri, Rust

- Reduces the burden of tracking context and progress separately across AI tools. Combines multiple AIs in one session with token and context tracking, parallel work, Git, issues and task management.

Mobile app

One Interview Question a Day Personalized interview practice

React Native, Python, FastAPI, Supabase, pgvector, Docker, Caddy · RAG

- Personalized daily interview practice using resume, role and experience context through RAG, with feedback on spoken and written responses.

Web service

Ildanom Study group management jsu.g0o0.com

TypeScript, Next.js, Supabase, PostgreSQL, Tailwind CSS

- Brings together weekly preparation records, scheduling and anonymous post-meeting feedback to reduce fragmented study group coordination.

Web learning tool

Tree Helper Visual tree learning tree.g0o0.com

TypeScript, Next.js, React, Tailwind CSS, Canvas API, Vercel

- Built and shared a web tool for constructing trees, inspecting results and practicing, addressing difficulties with drawing trees and checking answers in class. The professor still introduces and uses it in class. Received feedback that the number of students scoring 80 or above increased by over 30% in the semester it was used.

Practice service

Certification CBT Certification exam practice eip.g0o0.com

TypeScript, Next.js, React, Tailwind CSS, Vercel

- Built while studying for a certification exam and shared with an online community so others could practice and provide feedback.

Awards & Education

ICT Creativity Competition · Silver Award Aug 2018

Kookmin University Algorithm Competition · Encouragement Award Aug 2018

Kookmin University · Software 2019 - Feb 2026

Graduated · Admitted through the special-talent track